Result 1 (named relative homology image membership). [Topology foundations, module map range membership]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be a certified inclusion between pairs in X. Assume the chain square of Q commutes (`h`). Let $M = H(Q.\text{source})$. Let $N = H(Q.\text{target})$. Let $f = H(Q)$. Let $U = \text{range}(H(Q))$. Let x be an element of N. Then

$$x \text{ belongs to } U \text{ if and only if } (\text{there exists } y : M,\ f(y) = x)\,.$$

Adapted from the cited source.

Result 2 (named relative quotient maps commute). [Topology foundations, cokernel projection naturality]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be a certified inclusion between pairs in X. Assume the chain square of Q commutes (`h`). Let $A = C_*(Q.\text{source.ambient}; H)$. Let $B = C_*(Q.\text{target.ambient}; H)$. Let $C = C_*(Q.\text{source}; H)$. Let $D = C_*(Q.\text{target}; H)$. Let $f = C_*(Q)$. Let $p = \pi_{Q.\text{source}}$. Let $q = \pi_{Q.\text{target}}$. Let $g = C_*(Q.\mathit{ambient})$. Then

$$f \circ p = q \circ g.$$

Adapted from the cited source.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.